

OFFICIAL SENSITIVE

System Requirements Document — Integrated Command and Control Platform

DEF-SRD-2024-006

Issue 1.0 — RESTRICTED CIRCULATION

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Annex A — Functional Requirements

1 This annex defines the functional requirements for the Integrated Command and Control (ICC) Platform. All requirements are mandatory unless explicitly stated otherwise. Requirements shall be verified by the method indicated in the Verification Method attribute.

1.1 The system shall implement command and control functions that enable authorised operators to direct and monitor all assigned assets in real time.

1.1.1 The system shall display the current operational status of all assets in the common operating picture (COP) with a maximum update latency of 2 seconds from the time of status change at the asset.

1.1.2 The system shall provide a hierarchical command structure that mirrors the operational chain of command as defined in the current operational order.

1.1.2.1 The command hierarchy shall be configurable by an operator with the System Administrator role and shall take effect within 60 seconds of configuration being saved.

1.1.2.2 The system shall enforce that commands are only propagated downward through the command hierarchy and not laterally between units at the same echelon.

1.1.3 The system shall record all commands issued, acknowledged, and executed, together with the issuing operator identity, timestamp, and asset acknowledgement timestamp, in a tamper-evident command log.

1.1.3.1 The command log shall be stored in an append-only data structure and shall be replicated to the backup system within 5 seconds of each log entry being written.

1.1.3.1.1 The replication channel shall use mutual TLS authentication and shall be monitored for availability; any replication failure shall generate a P1 alert within 30 seconds.

1.1.3.1.1.1 The P1 alert shall be delivered simultaneously to the system duty officer display terminal, the operations room alarm panel, and the on-call engineer mobile device, and shall not be dismissible until acknowledged by a duty officer with an appropriate security clearance level.

1.2 The system shall aggregate situational awareness data from all connected sensor systems and present it in a fused, deduplicated common operating picture.

1.2.1 The system shall accept data feeds from the following sensor categories and shall process each within the latency tolerance specified:

1.2.2 Duplicate tracks arising from overlapping sensor coverage shall be fused within 500ms of the second observation being received, using the track fusion algorithm specified in the System Design Document.

1.2.3 The system shall permit an operator to manually correlate or de-correlate tracks and shall record the operator identity and timestamp of each manual correlation action.

1.3 The system shall provide a mission planning capability enabling authorised planners to create, edit, and distribute operational plans.

1.3.1 A mission plan shall contain at minimum: mission objective, execution timeline, asset assignments, rules of engagement reference, and designated communication channels.

1.3.2 The mission planning module shall enforce version control such that each saved revision of a plan creates an immutable record accessible to authorised reviewers.

1.3.2.1 The version history shall be accessible to any operator with at least Read-Only access to the plan and shall display the author, timestamp, and change summary for each revision.

1.3.2.1.1 Change summaries shall be mandatory for any revision that modifies asset assignments, rules of engagement references, or the execution timeline by more than 30 minutes in either direction.

1.4 The system shall manage communications between all nodes in the command network with priority-based queuing and guaranteed delivery for critical message types.

1.4.1 Critical messages (FLASH and IMMEDIATE precedence) shall be delivered within 5 seconds of transmission for all addressees within radio range or connected via primary data link.

1.4.2 The system shall provide an offline message store-and-forward capability for addressees that are temporarily out of communication, with automatic delivery upon reconnection.

Annex B — Interface Requirements

1 This annex defines the interface requirements for all external and internal interfaces of the ICC Platform. Interfaces are categorised as: External (systems outside programme boundary), Internal (between subsystems within the programme boundary), and Operator (human-machine interfaces).

1.1 The system shall implement external interfaces to all systems listed in the Interface Control Document (ICD-ICC-2024-001) using the protocols and data formats specified therein.

1.1.1 Each external interface shall have a defined service level agreement covering minimum bandwidth, maximum latency, and availability, as specified in the ICD.

1.1.2 The system shall monitor the health of each external interface and shall raise an alert to the duty operator if an interface falls below its defined service level thresholds.

1.1.2.1 Health monitoring data for external interfaces shall be retained for 30 days and shall be available to system administrators for capacity planning and incident investigation.

1.2 Internal interfaces between ICC Platform subsystems shall comply with the subsystem interface specifications defined in the System Design Document (SDD).

1.2.1 All internal interfaces shall use the platform-standard message bus (Apache Kafka v3.x) and shall conform to the canonical message schemas defined in the Schema Registry.

1.2.2 Each subsystem shall implement a health endpoint that returns a JSON status object within 200ms, conforming to the platform health-check schema.

1.2.2.1 The health endpoint shall be polled by the platform orchestration layer at least every 30 seconds and the results shall be surfaced in the platform operations dashboard.

1.3 The operator interface shall be designed for use in a high-workload, time-critical operational environment where errors may have significant consequences.

1.3.1 The interface shall apply the following human factors principles throughout:

1.3.2 Critical actions (issue command, approve mission plan, modify rules of engagement) shall require a mandatory confirmation step before execution. The confirmation dialogue shall state the action to be performed and its immediate consequences.

1.3.3 The interface shall provide keyboard shortcuts for all primary operator functions and shall make these shortcuts visible in the interface at all times.

1.4 All data exchanged with external systems shall use the formats specified in the ICD and shall conform to relevant NATO STANAG data standards where applicable.

1.4.1 Track data exchanged via the tactical data link interface shall conform to STANAG 5516 Link 16 message format.

1.4.2 Position reporting data shall conform to STANAG 4677 Friendly Force Tracking (FFT) message format.

1.4.2.1 Where a connected system cannot produce STANAG 4677 compliant output, the system shall implement a translation layer as specified in the relevant Legacy Interface Supplement.

Annex C — Performance Requirements

1 The performance requirements in this annex are binding service level requirements. Failure to meet any mandatory performance requirement constitutes a non-conformance and shall be raised as a defect in the project issue register.

1.1 The system shall meet the following end-to-end latency requirements for data flowing from sensor input to operator display.

1.1.1 Track data from all sensor inputs shall be processed and displayed on the COP within 2 seconds of receipt at the system ingress point, measured end-to-end under normal load conditions.

1.1.2 Command messages issued by an operator shall be transmitted to the target asset command receiver within 500ms of the operator pressing the 'Send' button, measured under normal RF conditions.

1.1.3 The mission planning module shall render a mission plan containing up to 500 waypoints and 200 asset assignments within 3 seconds of opening.

1.2 The system shall sustain the following minimum throughput levels under peak operational load, defined as 150 simultaneous operator sessions with all subsystems active.

1.2.1 Track processing throughput: minimum 10,000 track updates per minute.

1.2.2 Command message throughput: minimum 500 command messages per minute system-wide.

1.2.3 COP display refresh rate: minimum 1 Hz (one full screen update per second) for all connected operator terminals under peak load.

1.3 The system shall achieve a minimum operational availability of 99.9% measured over any 30-day operational period.

1.3.1 The system shall implement N+1 redundancy for all critical subsystems. The failure of any single component shall not result in loss of any primary operational capability.

1.3.2 Automatic failover between redundant components shall complete within 30 seconds of a primary component failure being detected by the health monitoring subsystem.

1.3.2.1 Failover events shall not result in loss of any command log entries or mission plan data. In-flight transactions interrupted by a failover shall be recovered using the transaction log within 60 seconds of failover completion.

1.4 The system shall be designed to accommodate the following capacity requirements for an initial 5-year operational life.

1.4.1 Maximum concurrent operator sessions: 200.

1.4.2 Maximum tracked entities in the common operating picture: 50,000 simultaneous tracks without performance degradation below the thresholds specified in 1.1.

1.4.3 Maximum stored mission plans: 10,000 plans with full version history retained for the operational life of the system.

1.4.3.1 Mission plan storage shall be designed to scale beyond 10,000 plans by adding storage capacity without architectural change, up to a maximum of 100,000 plans.

1.4.3.1.1 Scaling beyond 10,000 plans shall be achievable without system downtime using the storage expansion procedures defined in the Operations Manual.

1.4.3.1.1.1 The Operations Manual storage expansion procedure shall be tested and validated as part of the pre-deployment acceptance test, with evidence of a successful test retained in the project quality records.